Product Sum

AlgoExpert · Easy · Recursion

The problem

- A "special" array holds ints or nested special arrays; each nested array's sum is multiplied by its depth (outermost = 1).

Intuition

- A nested array's sum is multiplied by its depth, which is naturally recursive.

Brute force vs optimal

- Only real approach: recursive descent, O(n) total elements
- Mistake to avoid: multiplying by depth too early or twice
- Iterative version uses a stack of (list, depth)

Approach

- 1. Sum the elements at this level
- 2. For a nested list, recurse with depth plus one
- 3. Multiply the level sum by the depth
- 4. Return it

Worked example

```
[5, 2, [7, -1], 3, [6, [-13, 8], 4]]
```

```
depth 1: 5 + 2 + 3 = 10 (+ nested)
```

Depth 1: add plain numbers 5, 2, 3

Worked example

```
[5, 2, [7, −1], 3, [6, [−13, 8], 4]]
```

```
[7, −1] at depth 2: (7 − 1) × 2 = 12
```

Nested [7, −1]: sum 6 × depth 2 = 12

Worked example

```
[5, 2, [7, −1], 3, [6, [−13, 8], 4]]
```

```
[−13, 8] at depth 3: (−5) × 3 = −15
```

Innermost [−13, 8]: −5 × 3 = −15

Worked example

`[5, 2, [7, −1], 3, [6, [−13, 8], 4]]`

`[6, [−13, 8], 4]` at depth 2: $(6 - 15 + 4) \times 2 = -10$

Middle list: $(6 + (-15) + 4) \times 2 = -10$

Worked example

```
[5, 2, [7, -1], 3, [6, [-13, 8], 4]]
```

```
total: (5 + 2 + 12 + 3 − 10) × 1 = 12
```

Answer: 12

Java solution

```java
public class ProductSum {
    public static int productSum(List<Object> array) { return productSum(array, 1); }

    @SuppressWarnings("unchecked")
    private static int productSum(List<Object> array, int depth) {
        int sum = 0;
        for (Object o : array) sum += (o instanceof List) ? productSum((List<Object>) o, depth + 1) : (Integer) o;
        return sum * depth;
    }
}
```

Complexity

- Time: O(n) total elements
- Space: O(d) depth

Edge cases & pitfalls

- Empty nested lists contribute 0
- Multiply nested results by their own depth only
- Negative numbers

Interview tips & follow-ups

- Clarify the input type (Object list in Java)
- Follow-up: weight inversely by depth

Recap

- A nested array's sum is multiplied by its depth, which is naturally recursive.
- Complexity: O(n) total elements